

Introduction to Selenium

Web Application Automation Testing

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Subject: Software Testing



What is Selenium?

Selenium is an open-source framework used for automating web applications.

- It automates actions performed in a web browser.
- It is mainly used for software testing.
- It can perform tasks such as clicking, typing, selecting, and navigation.
- It supports multiple browsers and programming languages.
- Selenium is widely used for functional and regression testing.



Example

Selenium can automatically test whether a website's login page is working correctly.

Why Do We Need Selenium?

Manual testing can be repetitive and time-consuming.

Selenium helps to:

- ✓ Automate repetitive test cases
- ✓ Save testing time
- ✓ Reduce human errors
- ✓ Perform repeated tests quickly
- ✓ Test websites on different browsers
- ✓ Support regression testing
- ✓ Improve testing efficiency

Example: Instead of manually testing login 100 times, Selenium can perform the test automatically.

Key Features of Selenium



Open Source

Free to use



Cross-Browser Support

Chrome, Firefox,
Edge, Safari



Cross-Platform

Windows, Linux,
macOS



Multiple Languages

Java, Python, C#,
JavaScript, etc.



Web Automation

Automates browser actions



Parallel Testing

Tests can be executed
simultaneously



Integration

Works with testing
frameworks and CI/CD tools

Selenium Components

Selenium consists of three important components:



Selenium WebDriver

- Automates browsers using programming languages.
- Most commonly used Selenium component.



Selenium IDE

- Browser extension used to record and replay tests.
- Useful for beginners and simple automation.



Selenium Grid

- Allows tests to run on multiple machines and browsers.
- Useful for parallel testing.

Note: Selenium RC was an older component and is no longer used in modern Selenium.

Selenium WebDriver

WebDriver is the main component used for browser automation.

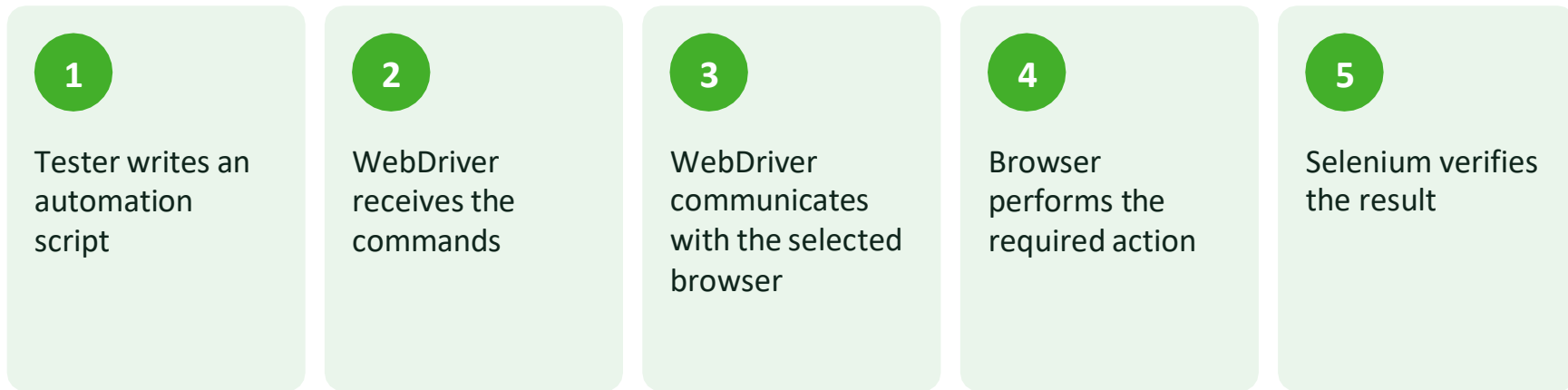
It allows a test program to communicate directly with a browser.



For example:

- 1**
Python script sends a command
- 2**
WebDriver communicates with Chrome
- 3**
Chrome performs the requested action
- 4**
Result is returned to the test

How Selenium Works



Example:



Simple Selenium Example

Python + Selenium

```
from selenium import webdriver
driver = webdriver.Chrome()
driver.get("https://www.google.com")
print(driver.title)
driver.quit()
```

What does this code do?

- 1 Opens Google Chrome
- 2 Opens Google website
- 3 Prints the page title
- 4 Closes the browser

Advantages of Selenium



Free and open source



Easy to integrate with programming languages



Supports multiple browsers



Supports multiple operating systems



Reduces manual testing effort



Suitable for regression testing



Large community and learning resources



Supports parallel test execution

Limitations of Selenium



Mainly designed for web applications



Requires programming knowledge for advanced automation



Does not directly automate desktop applications



Does not directly automate mobile apps



Reporting usually requires additional tools/frameworks



Test maintenance can be required when websites change

Selenium vs Manual Testing

Manual Testing	Selenium Automation
Performed by a tester	Performed by automation scripts
Time-consuming for repetitive tasks	Faster for repetitive tasks
More chance of human error	Consistent execution
Difficult to repeat frequently	Tests can be repeated easily
Good for exploratory testing	Good for regression testing

Conclusion: Selenium does not completely replace manual testing. Both approaches are useful for different testing situations.

Applications of Selenium

Selenium can be used for:



Functional Testing



Regression Testing



**Cross-Browser
Testing**



Smoke Testing



End-to-End Testing



**Web Application
Testing**



**Automated Test
Execution**

Selenium with Other Tools

Selenium can be integrated with various tools and frameworks.



Programming Languages

Python

Java

C#

JavaScript



Testing Frameworks

TestNG

JUnit

PyTest



Other Tools

Jenkins

Maven

Git

Selenium Grid

This allows Selenium to become part of an automated testing and CI/CD workflow.

Real-Life Example

Testing an E-Commerce Website — Selenium can automatically test:

1 Open the shopping website

2 Login to the account

3 Search for a product

4 Select the product

5 Add it to the cart

6 Verify the cart

7 Logout

This entire process can be automated instead of manually repeating it for every test cycle.

Conclusion: Key Takeaways

1

Selenium is an open-source web automation framework.

2

It is widely used for automated web application testing.

3

WebDriver is the main component for browser automation.

4

Selenium supports multiple browsers, operating systems, and programming languages.

5

It helps reduce repetitive manual testing effort.

6

It is especially useful for functional, regression, and cross-browser testing.

Thank You | Any Questions?